to provide coke to iron and steel industries. By-products of this coal can be converted into different chemicals which are used to make paint, nylon, and many other items.

Anthracite

Anthracite is the highest grade coal. It has a very light weight and the highest heat content. Anthracite coal is very hard, deep black and shiny. It contains 86-97% carbon and has a heating value slightly higher than bituminous coal. It burns longer with more heat and less dust.

Activity 4

In an outline map of India mark the places where coal mines are found. Also identify the type of coal found in those areas.

15.4.3 Uses of Coal

- Coal is used to generate heat and electricity.
- It is used to make derivatives of silicon which are used to make lubricants, water repellents, resins, cosmetics, hair shampoos, and toothpaste.
- Activated charcoal is used to make face packs and cosmetics.
- Coal is used to make paper.
- Coal helps to create alumina refineries.
- Carbon fibre which is an extremely strong but light weight material is used in construction, mountain bikes, and tennis rackets.



Figure 15.13 Uses of Coal

- Activated carbon, used in filters for water and air purification and in kidney dialysis machines is obtained from coal.

15.4.4 Products obtained from coal

Coal when heated in the absence of air does not burn but produces many by-products. This process of heating coal in the absence of air is called destructive distillation of coal. The destructive distillation of coal can be carried out in the laboratories. The apparatus is as shown in Figure 15.13.

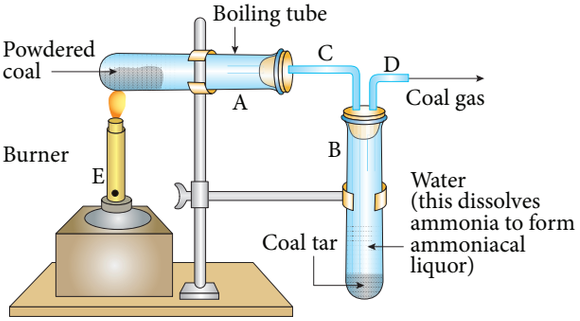


Figure 15.13 Destructive Distillation

Finely powdered coal is taken in a test tube and heated. At a particular temperature coal breaks down to produce coke, coal tar, ammonia and coal gas. Coal tar is deposited at the bottom of the second test tube and coal gas escapes out through the side tube. Ammonia produced is absorbed in the water, forming ammonium hydroxide. Finally a black residue called coke is left in the tube.

Thousands of different products have coal or coal by-products as their components. Some of them are soap, aspirins (tablet), solvents, dyes, plastics, and fibres, such as rayon and nylon. The main by products obtained during destructive distillation are coke, coal tar, ammonia and coal gas.

Coke

Coke contains 98% carbon. It is a porous, black and the purest form of coal. It is a good fuel and burns without smoke. It is largely used as a reducing agent in the

8th_Science_Unit-15_EM.indd 172 05-03-2020 14:04:01